

ETF3231/5231 Individual Assignment 4

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Your aim in the first part of the assignment is to build an ARIMA model and use this to forecast. Recall the first step in ARIMA modelling is to stabilize the variance of your data. If you decided that your data required a transformation in the previous assignments you will be required to use the same transformation for what follows (unless you have a reason to change your mind - please do so if you think it is necessary).

Question 1

Visually inspect your data and decide on the transformation and what differencing is required to achieve stationarity. Plot the data at every step and comment on each plot justifying your actions. (No more than 50 words per plot). (6 marks)

```
my_series %>%  
  autoplot(y) +  
  labs(title = "Queensland Takeaway Food Services Turnover",  
        x = "Month", y = "Turnover ($ Millions)")
```

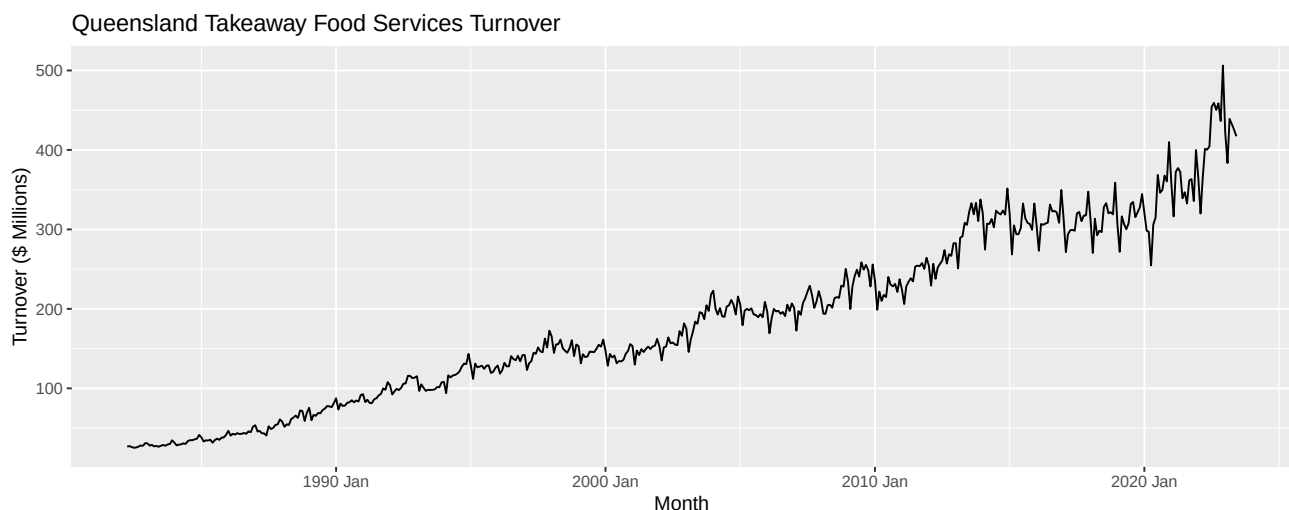


Figure 1: Original time plot for Queensland takeaway food services turnover.

This series in Figure 1 shows monthly Queensland takeaway food services turnover in millions of dollars from April 1982 to June 2023. The series has a strong upward trend, monthly seasonality and increasing variation as the level rises. Since the data are monthly, $m = 12$, so seasonal differencing may be needed.

```
lambda_tbl <- my_series %>%
  features(y, guerrero)

lambda <- lambda_tbl$lambda_guerrero

lambda_tbl %>%
  mutate(lambda_guerrero = round(lambda_guerrero, 4)) %>%
  rename(`Guerrero lambda` = lambda_guerrero) %>%
  kable(digits = 4)
```

Table 1: Guerrero lambda for the original series.

Guerrero lambda
0.0595

```
p1 <- my_series %>%
  autoplot(sqrt(y)) +
  labs(title = "Square-root", x = "Month", y = "sqrt(Turnover)")

p2 <- my_series %>%
  autoplot(y^(1/3)) +
  labs(title = "Cube-root", x = "Month", y = "cube-root(Turnover)")

p3 <- my_series %>%
  autoplot(log(y)) +
  labs(title = "Log", x = "Month", y = "log(Turnover)")

p4 <- my_series %>%
  autoplot(box_cox(y, lambda)) +
  labs(title = "Box-Cox using Guerrero lambda", x = "Month", y = "Box-Cox(Turnover)")

p1 / p2 / p3 / p4
```

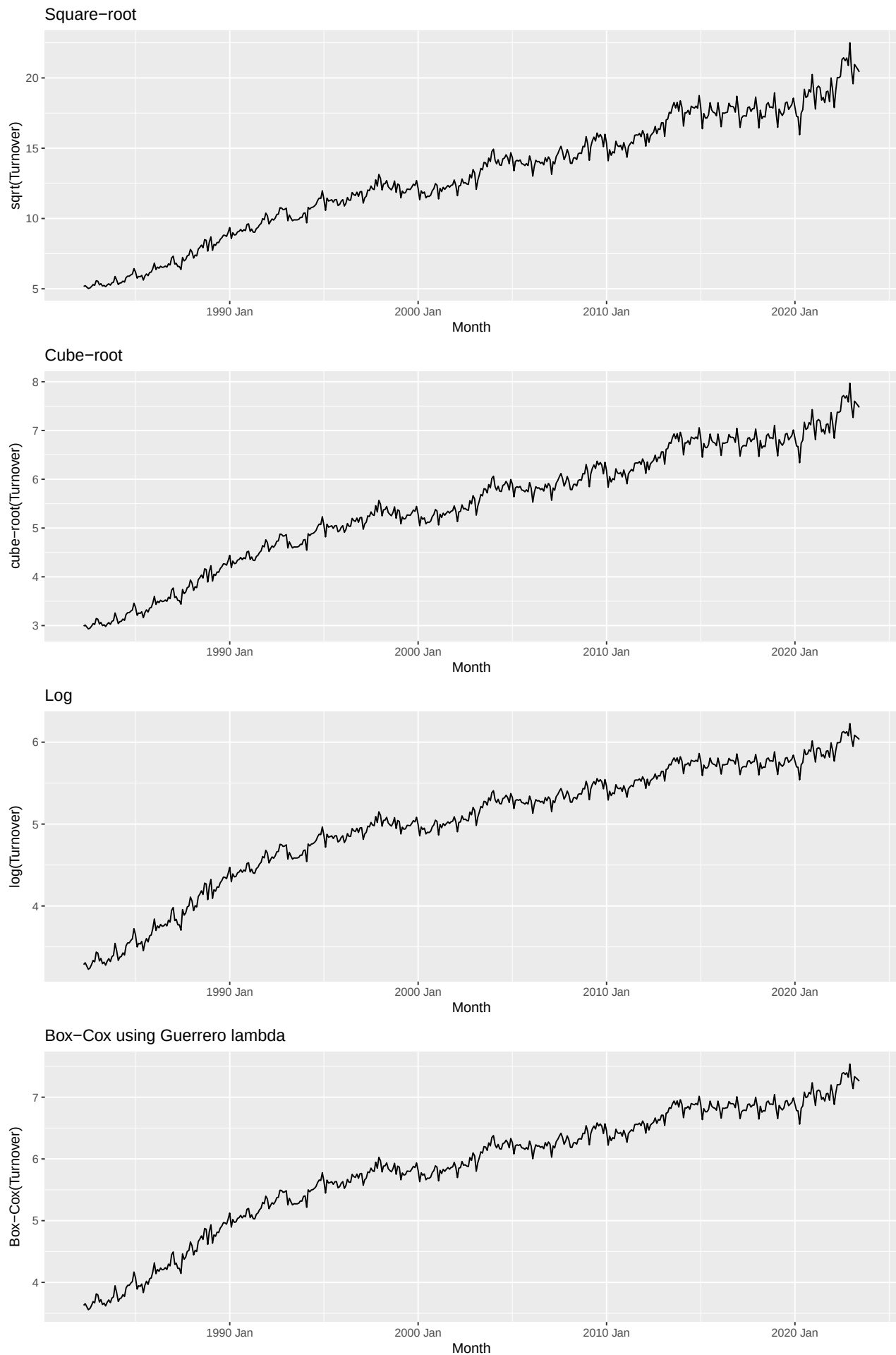


Figure 2: Comparison of transformations for Queensland takeaway food services turnover.

Figure 2 compares possible transformations. The original series shows increasing variation as the level rises. Square-root and cube-root transformations reduce the variance, but the plots still show increasing variation at higher levels. The log and Box-Cox transformations stabilise the spread best. Since Guerrero's lambda is close to zero, I use $\log(y)$.

```
my_series %>%  
  autoplot(log(y)) +  
  labs(title = "Log transformed Queensland takeaway food services turnover",  
        x = "Month", y = "log(Turnover)")
```



Figure 3: Log transformed turnover series.

Figure 3 still shows trend and monthly seasonality after transformation, so differencing is needed. The log transformation is kept because the spread is more stable than the original scale.

```
my_series %>%  
  autoplot(difference(log(y), 12)) +  
  labs(title = "Seasonally differenced log turnover",  
        x = "Month", y = "Seasonal difference of log(Turnover)")
```

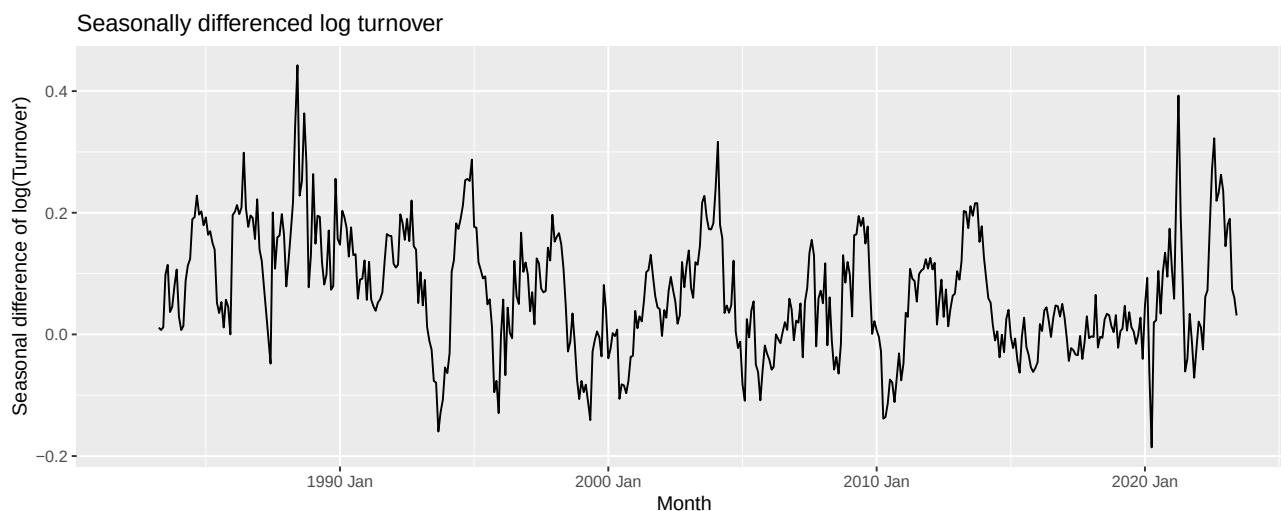


Figure 4: Seasonally differenced log turnover series.

Figure 4 shows that seasonal differencing removes much of the yearly pattern, but the series still has slow movement over time. Therefore, a further first difference is needed.

```
my_series %>%
  autoplot(difference(log(y), 12) %>% difference()) +
  labs(title = "Seasonally and first differenced log turnover",
       x = "Month", y = "Differenced log(Turnover)")
```

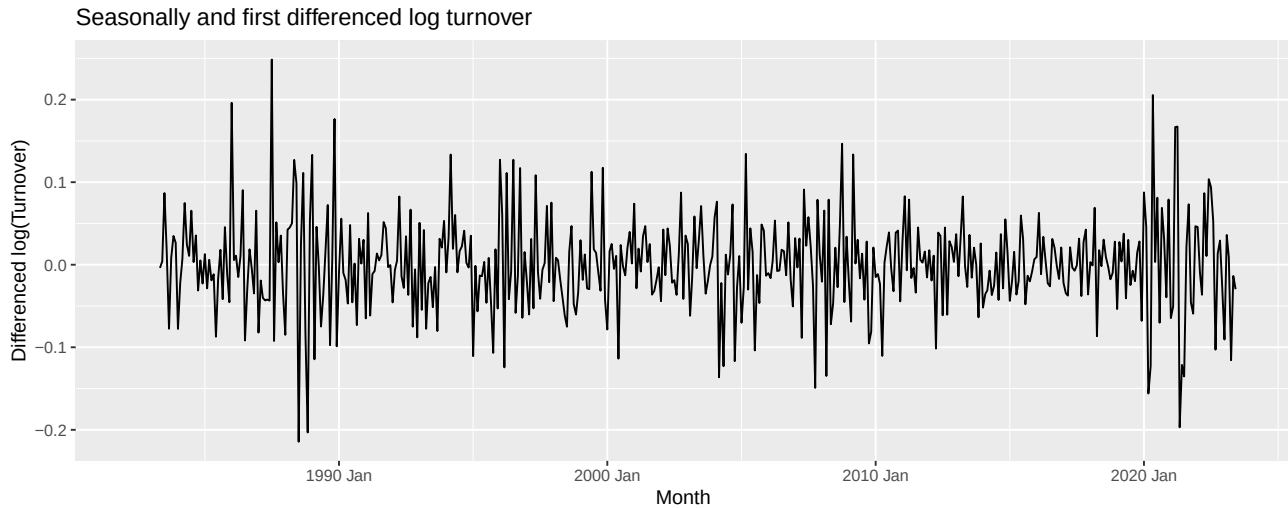


Figure 5: Seasonally and first differenced log turnover series.

Figure 5 fluctuates around zero with a more stable spread and no obvious remaining trend. I therefore use $\log(y)$ with one seasonal difference and one first difference, so the ARIMA models below use $D = 1$, $d = 1$, and $m = 12$.

```
seasonal_diff_tbl <- my_series %>%
  features(log(y), unitroot_nsdiffs)

first_diff_tbl <- my_series %>%
  features(difference(log(y), 12), unitroot_ndiffs)

bind_cols(seasonal_diff_tbl, first_diff_tbl) %>%
  kable()
```

Table 2: Unit root checks used as supporting evidence for differencing.

nsdiffs	ndiffs
1	1

Table 2 shows supporting unit root checks for the differencing decision. The test output suggests one seasonal difference and one ordinary difference, which agrees with my decision above.

Question 2

Plot the ACF and PACF of the stationary data. Reading from these choose an appropriate ARIMA model. Make sure you justify your choice. (No more than 70 words in total – do not revise the theory – describe what you see in your plots and decide what ARIMA orders may be appropriate. Also note that it is highly likely that the ACF and PACF plots will be very messy. Do the best you can). (6 marks)

```
my_series %>%  
  gg_tsdisplay(difference(log(y), 12) %>% difference(),  
    plot_type = "partial",  
    lag_max = 48) +  
  labs(title = "Stationary transformed series")
```

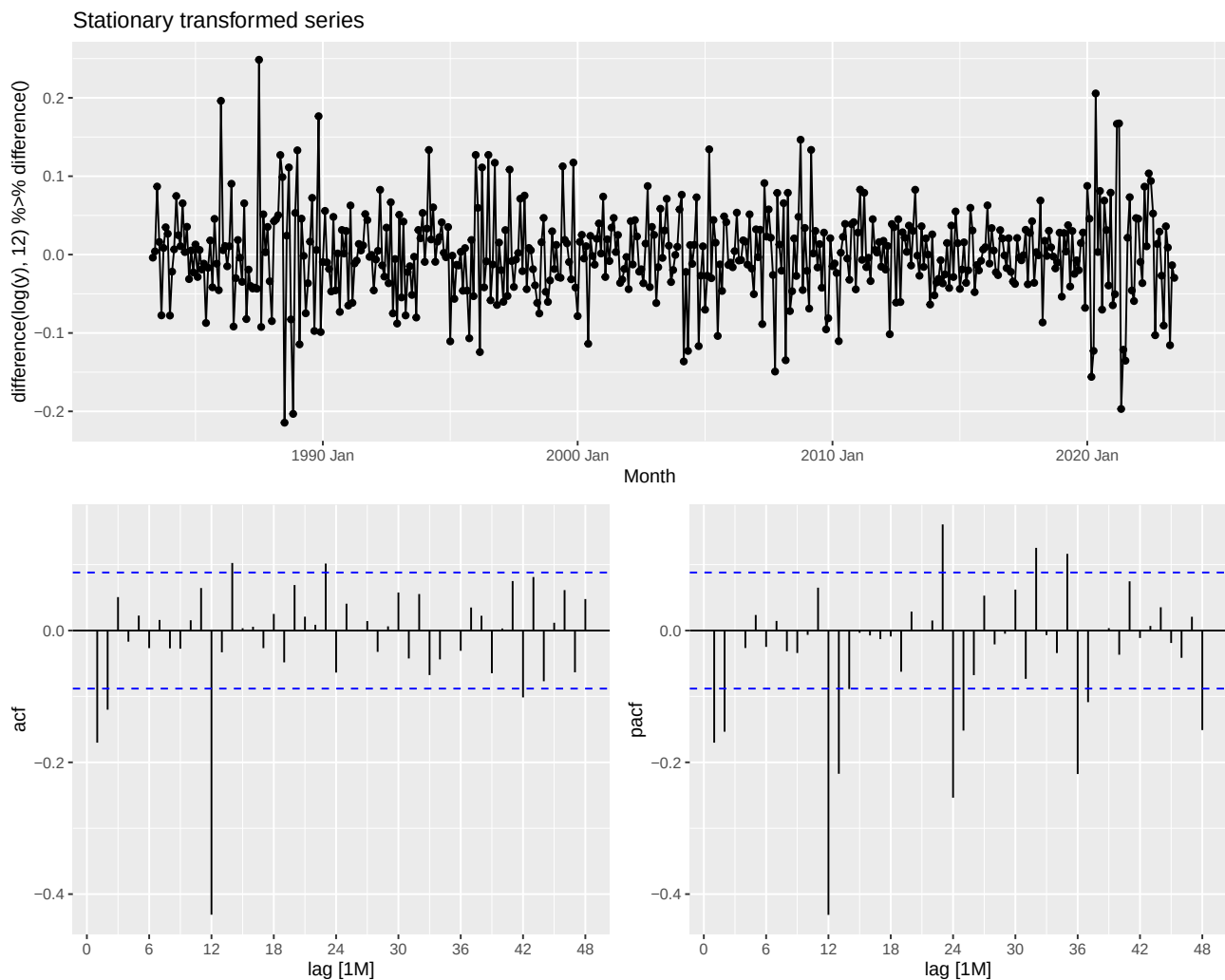


Figure 6: ACF and PACF of the stationary transformed series.

At non-seasonal lags, both ACF and PACF show early spikes up to about lag 2, suggesting possible MA(2) and AR(2). I use the PACF-based AR(2) as the initial choice because it is clearer. At seasonal lags, the ACF cuts off after lag 12, while the PACF gradually decays. I therefore choose seasonal MA(1), giving $\text{ARIMA}(2,1,0)(0,1,1)_{12}$.

Question 3

Check the whiteness of the residuals from the fitted ARIMA model. Based on these evaluate and if necessary review the ARIMA model specified in Q2. (No more than 50 words). (4 marks)

```
q3_fit <- my_series %>%  
  model(q2_arima = ARIMA(log(y) ~ pdq(2, 1, 0) + PDQ(0, 1, 1)))  
  
report(q3_fit)
```

```
Series: y  
Model: ARIMA(2,1,0)(0,1,1)[12]  
Transformation: log(y)
```

```
Coefficients:  
          ar1      ar2      sma1  
-0.2734 -0.1608 -0.8529  
s.e.    0.0450  0.0449  0.0272
```

```
sigma^2 estimated as 0.001857: log likelihood=825.42  
AIC=-1642.84 AICc=-1642.76 BIC=-1626.13
```

```
q3_fit %>%  
  gg_tsresiduals(lag = 48)
```

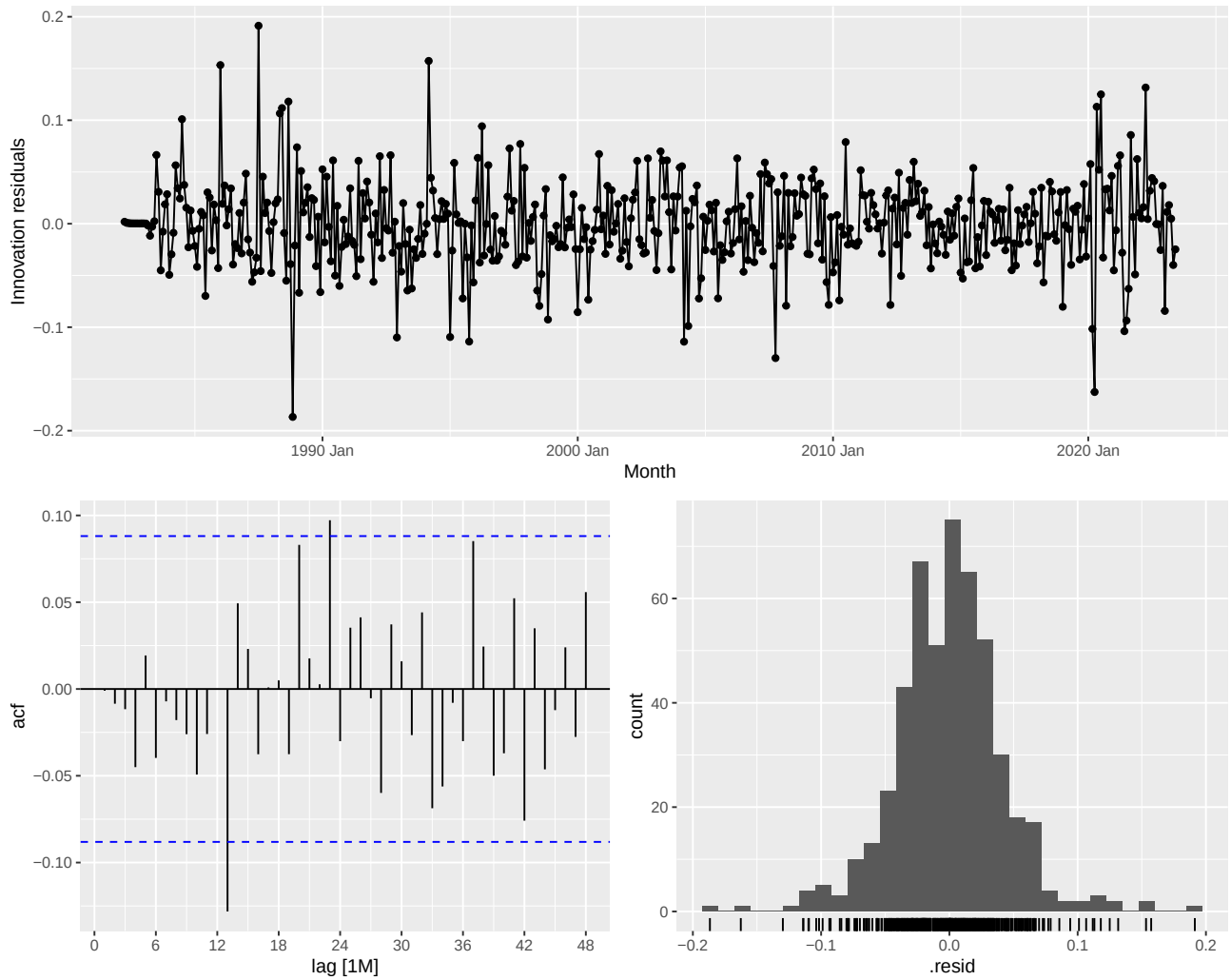


Figure 7: Residual diagnostics for the ARIMA(2,1,0)(0,1,1)[12] model.

```
q3_fit %>%
  augment() %>%
  features(.innov, ljung_box, lag = 24, dof = 3) %>%
  kable(digits = 3)
```

Table 3: Ljung-Box test for ARIMA residual whiteness.

.model	lb_stat	lb_pvalue
q2_arima	24.707	0.26

The residual plot shows no trend or seasonality, but two ACF spikes remain. I use Ljung-Box with lag 24 and dof 3, from two AR terms and one seasonal MA term. Since $p = 0.260 > 0.05$, I fail to reject $H_0 : \rho_1 = \dots = \rho_{24} = 0$ jointly at 5%; residuals are acceptable.

Question 4

Consider three (up to five if you think you need them) alternative ARIMA models based on your choice in Q2 and Q3. (Very briefly justify each choice with no more than 1 or 2 lines each). Use information criteria to choose the best model you have considered so far. (Very briefly justify each choice with no more than 1 or 2 lines each). (6 marks)

```
q4_fit <- my_series %>%
  model(arima210011 = ARIMA(log(y) ~ pdq(2,1,0) + PDQ(0,1,1)),
        arima012011 = ARIMA(log(y) ~ pdq(0,1,2) + PDQ(0,1,1)),
        arima210110 = ARIMA(log(y) ~ pdq(2,1,0) + PDQ(1,1,0)),
        arima012110 = ARIMA(log(y) ~ pdq(0,1,2) + PDQ(1,1,0)))
```

The following models are considered because the ACF and PACF in Q2 suggested two possible non-seasonal structures and two possible seasonal structures.

- **arima210011**: This is the initial model from Q2. It uses the PACF-based non-seasonal AR(2) and the clearer seasonal ACF cut-off after lag 12, suggesting seasonal MA(1).
- **arima012011**: This checks the ACF-based non-seasonal MA(2) alternative, while keeping the seasonal MA(1) term suggested by the seasonal ACF.
- **arima210110**: This keeps the PACF-based non-seasonal AR(2), but checks the seasonal PACF alternative by using seasonal AR(1) instead of seasonal MA(1).
- **arima012110**: This combines the ACF-based non-seasonal MA(2) alternative with the seasonal PACF-based AR(1) alternative.

```
q4_fit %>%
  glance() %>%
  arrange(AICc) %>%
  select(.model, sigma2, log_lik, AIC, AICc, BIC) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  kable()
```

Table 4: Information criteria for manually selected ARIMA models.

.model	sigma2	log_lik	AIC	AICc	BIC
arima210011	0.002	825.421	-1642.843	-1642.759	-1626.131
arima012011	0.002	825.223	-1642.446	-1642.362	-1625.734
arima210110	0.002	762.064	-1516.129	-1516.045	-1499.417
arima012110	0.002	761.854	-1515.709	-1515.625	-1498.997

Based on Table 4, **arima210011**, ARIMA(2,1,0)(0,1,1)₁₂, has the lowest AICc, so it is the best model considered so far. **arima012011** has a very similar AICc, so the ACF-based non-seasonal MA(2) alternative is also plausible. However, **arima210110** and **arima012110** have much higher AICc values. This suggests that the seasonal MA(1) structure is more appropriate than the seasonal AR(1) structure, so keeping (Q = 1) is important.

Question 5

Let the `ARIMA()` function choose a model. If you need to, make `ARIMA()` search harder exploring all possible options within the function. Perform a residual diagnostics analysis of the chosen model. How does this compare with your chosen model from Q4? (No more than 100 words). (6 marks)

```
q5_fit <- my_series %>%  
  model(  
    manual = ARIMA(log(y) ~ pdq(2,1,0) + PDQ(0,1,1)),  
    auto = ARIMA(log(y),  
                  stepwise = FALSE,  
                  approximation = FALSE))
```

```
q5_fit %>%  
  select(auto) %>%  
  report()
```

Series: y
Model: ARIMA(2,1,0)(0,1,1)[12]
Transformation: log(y)

Coefficients:
 ar1 ar2 sma1
 -0.2734 -0.1608 -0.8529
s.e. 0.0450 0.0449 0.0272

sigma^2 estimated as 0.001857: log likelihood=825.42
AIC=-1642.84 AICc=-1642.76 BIC=-1626.13

```
q5_fit %>%  
  glance() %>%  
  select(.model, sigma2, log_lik, AIC, AICc, BIC) %>%  
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%  
  kable()
```

Table 5: Information criteria comparison between the manual and automatic ARIMA models.

.model	sigma2	log_lik	AIC	AICc	BIC
manual	0.002	825.421	-1642.843	-1642.759	-1626.131
auto	0.002	825.421	-1642.843	-1642.759	-1626.131

```
q5_fit %>%  
  tidy() %>%  
  select(.model, term, estimate, std.error) %>%  
  mutate(across(where(is.numeric), ~ round(.x, 4))) %>%  
  kable()
```

Table 6: Parameter comparison between the manual and automatic ARIMA models.

.model	term	estimate	std.error
manual	ar1	-0.2734	0.0450
manual	ar2	-0.1608	0.0449
manual	sma1	-0.8529	0.0272
auto	ar1	-0.2734	0.0450
auto	ar2	-0.1608	0.0449
auto	sma1	-0.8529	0.0272

```
q5_fit %>%
  select(auto) %>%
  gg_tsresiduals(lag = 48)
```

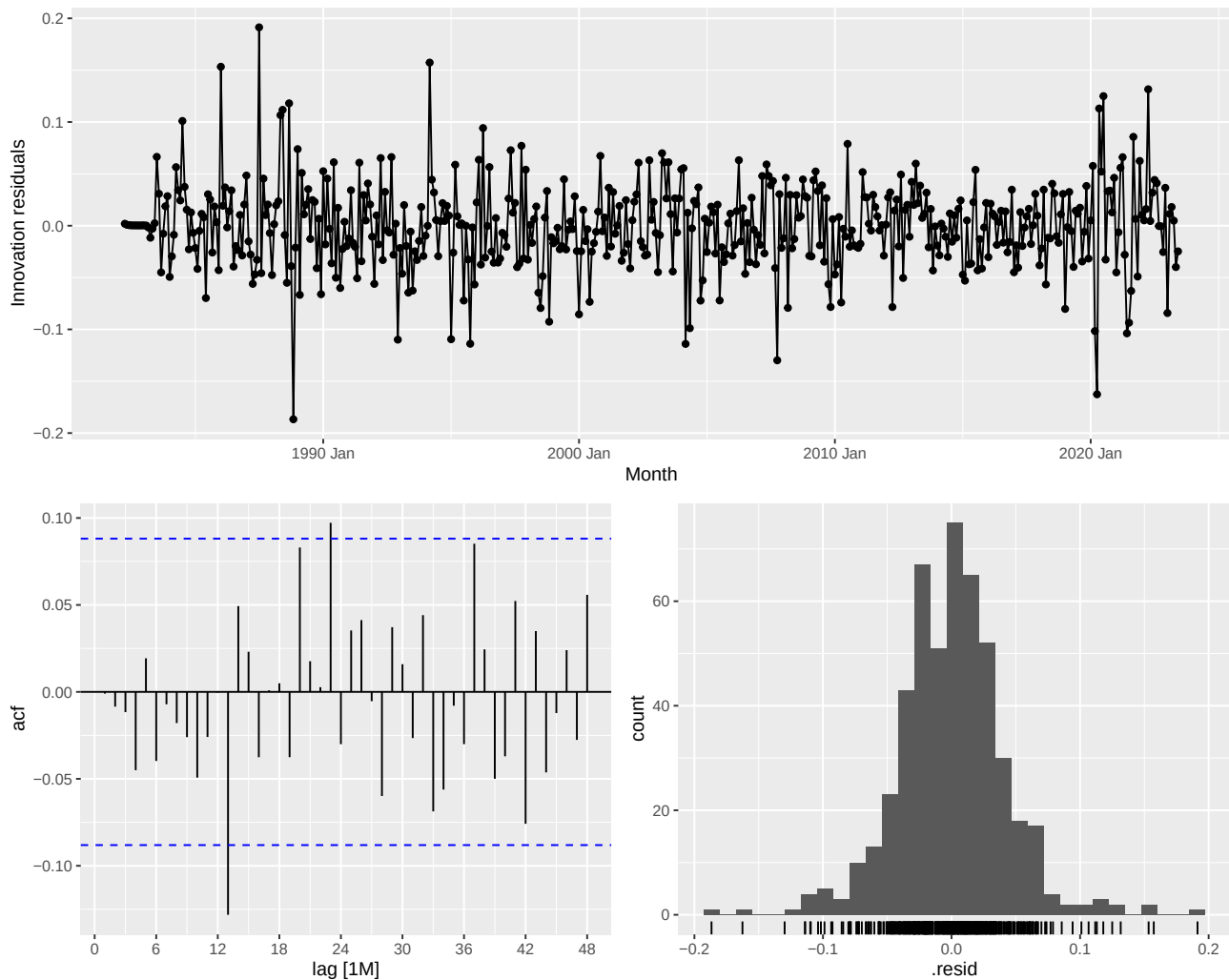


Figure 8: Residual diagnostics for the automatically selected ARIMA model.

```
q5_fit %>%
  select(auto) %>%
  augment() %>%
  features(.innov, ljung_box, lag = 24, dof = 3) %>%
  kable(digits = 3)
```

Table 7: Ljung-Box test for residual whiteness of the automatically selected ARIMA model.

.model	lb_stat	lb_pvalue
auto	24.707	0.26

The full search with `stepwise = FALSE` and `approximation = FALSE` selected the same ARIMA(2,1,0)(0,1,1)₁₂ model as my manual choice. Since there is no separate automatic model to compare, I confirm equivalence using orders, estimates in Table 6, AICc in Table 5, and diagnostics in Figure 8 and Table 7. This supports Q4. Residuals show no trend or seasonality, but two ACF spikes remain. Using lag 24 and dof 3 from two AR terms and one seasonal MA term, I test $H_0 : \rho_1 = \dots = \rho_{24} = 0$ jointly. Since $p = 0.260 > 0.05$, I fail to reject H_0 at 5%.

This identical-model situation was discussed with teaching staff on the discussion forum. Please see Ed post #201.

Question 6

Remember that you cannot use information criteria to compare between models with different orders of differencing. If necessary use an appropriate test set to choose the ARIMA model you want to use for forecasting. Which model have you selected and why? (No more than 50 words). (4 marks)

Since I am not comparing ARIMA models with different differencing orders, a test set is not needed here. All considered models use $\log(y)$, $d = 1$ and $D = 1$. I therefore select $\text{ARIMA}(2,1,0)(0,1,1)_{12}$ because it has the lowest AICc, acceptable residual diagnostics, and matches the automatic ARIMA result.

This approach was confirmed with teaching staff on the discussion forum. Please see Ed post #201.

Question 7

Generate and plot forecasts and forecast intervals from your chosen ARIMA model for two years following the end of your sample. Comment on these. (No more than 50 words). (3 marks)

```
q7_fc <- q5_fit %>%
  select(auto) %>%
  forecast(h = "2 years")

q7_fc %>%
  autoplot(my_series, level = c(80, 95)) +
  labs(title = "Two-year forecast for Queensland takeaway food services turnover",
       x = "Month", y = "Turnover ($ Millions)")
```

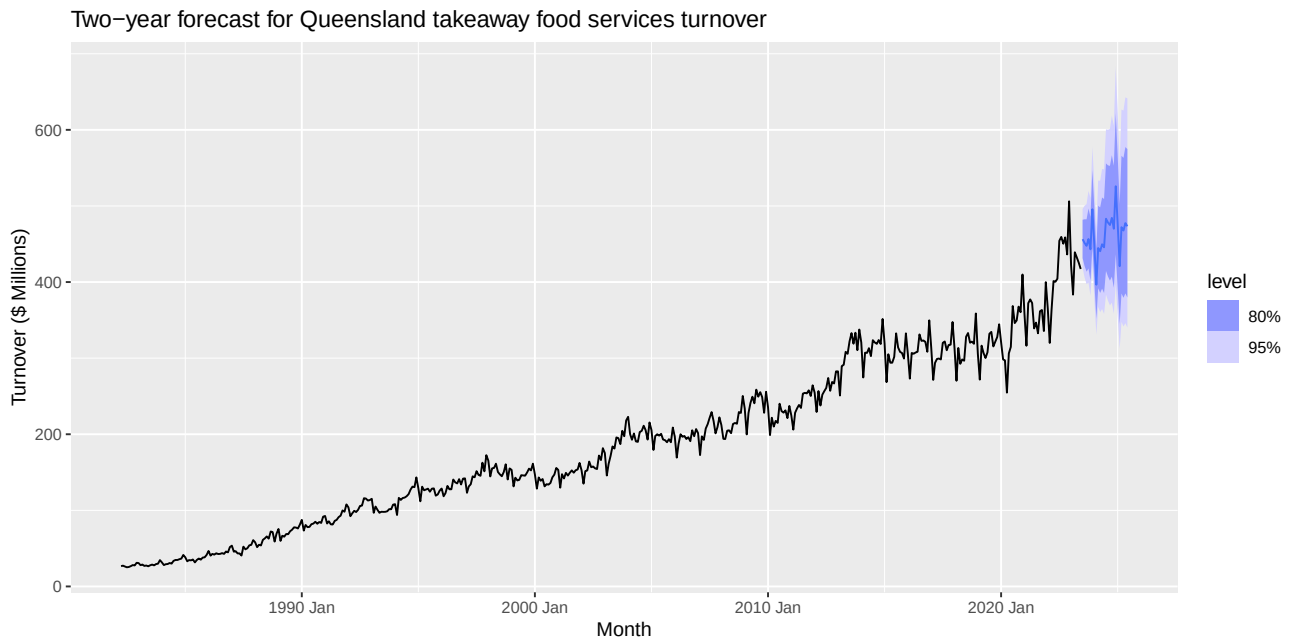


Figure 9: Two-year forecasts from the selected ARIMA(2,1,0)(0,1,1)[12] model.

```
q7_fc %>%
  autoplot(my_series %>% filter(year(Month) >= 2018),
          level = c(80, 95)) +
  labs(title = "Zoomed two-year forecast for Queensland takeaway food services turnover",
       x = "Month",
       y = "Turnover ($ Millions)")
```

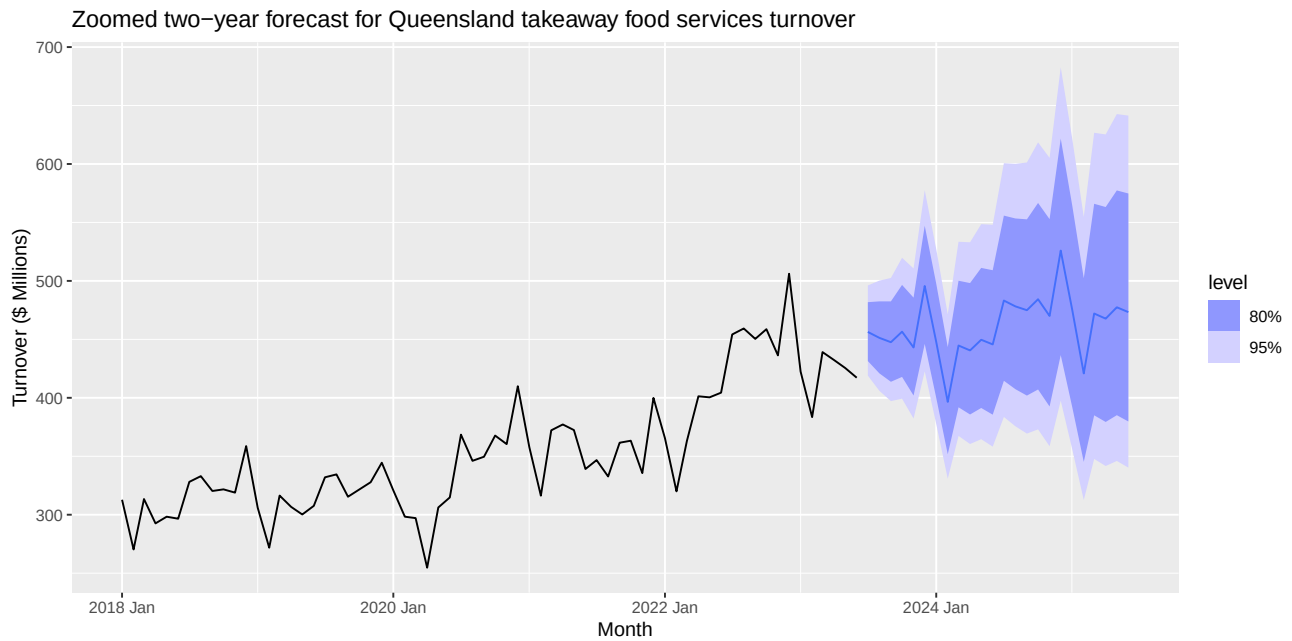


Figure 10: Zoomed two-year forecasts from the selected $ARIMA(2,1,0)(0,1,1)[12]$ model.

The forecasts continue the recent upward movement and retain monthly seasonal variation. The model captures the strong increase after 2020 and extends this higher level into the forecast period. The prediction intervals widen because uncertainty accumulates in the differenced ARIMA model, and two residual ACF spikes suggest some remaining uncertainty.

Question 8

You have now considered several modelling frameworks and built several models for your data set. In this part of the assignment you will evaluate these.

Update your data from the ABS website <https://www.abs.gov.au/statistics/industry/retail-and-wholesale-trade/retail-trade-australia>. Explore your updated time series and comment on the series movement through and post the COVID-19 pandemic. Provide any necessary plots to support your analysis. Some States and Industries are unfortunately affected more than others. (6 marks)

```
updated_series <- read_excel(
  "8501011.xlsx", sheet = "Data1", skip = 9) %>%
  select(Month = 1, y = A3349718A) %>%
  mutate(Month = yearmonth(as.Date(Month)),
         y = as.numeric(y)) %>%
  as_tsibble(index = Month)

monthly_counts <- updated_series %>%
  as_tibble() %>%
  count(Month)

updated_summary <- monthly_counts %>%
  summarise(
    `Start month` = min(Month),
    `End month` = max(Month),
    `One observation for each month` =
      if_else(sum(n) == ((year(max(Month)) - year(min(Month))) * 12 +
        (month(max(Month)) - month(min(Month))) + 1) &
        min(n) == 1 & max(n) == 1,
        "Yes", "No"))

updated_summary %>%
  kable()
```

Table 8: Coverage check for the updated Queensland takeaway food services turnover series.

Start month	End month	One observation for each month
1982 Apr	2025 Jun	Yes

```
updated_series %>%
  autoplot(y) +
  labs(title = "Updated Queensland takeaway food services turnover",
       x = "Month", y = "Turnover ($ Millions)")
```

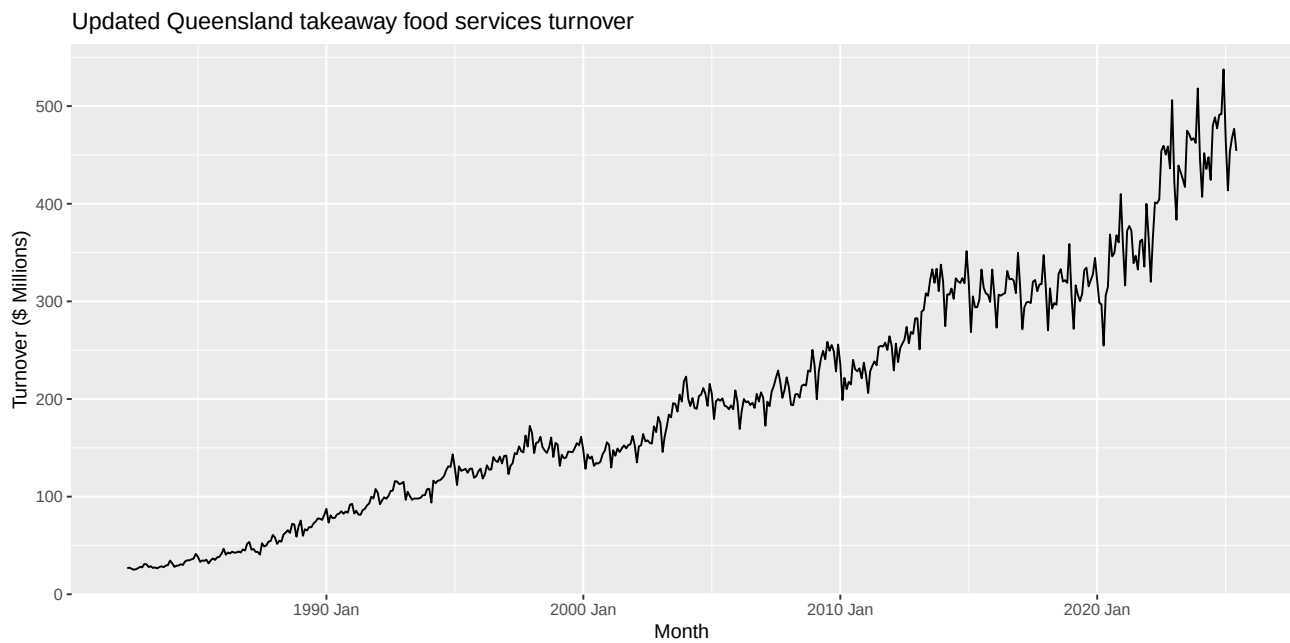



Figure 11: Updated Queensland takeaway food services turnover series.

```
updated_series %>%
  filter(year(Month) >= 2018) %>%
  autoplot(y) +
  geom_vline(xintercept = yearmonth("2020 Jan"),
             linetype = "dashed") +
  labs(
    title = "COVID and post-COVID movement in Queensland takeaway food services",
    subtitle = "Dashed line marks the January 2020 COVID-19 public health emergency reference point",
    x = "Month",
    y = "Turnover ($ Millions)")
```

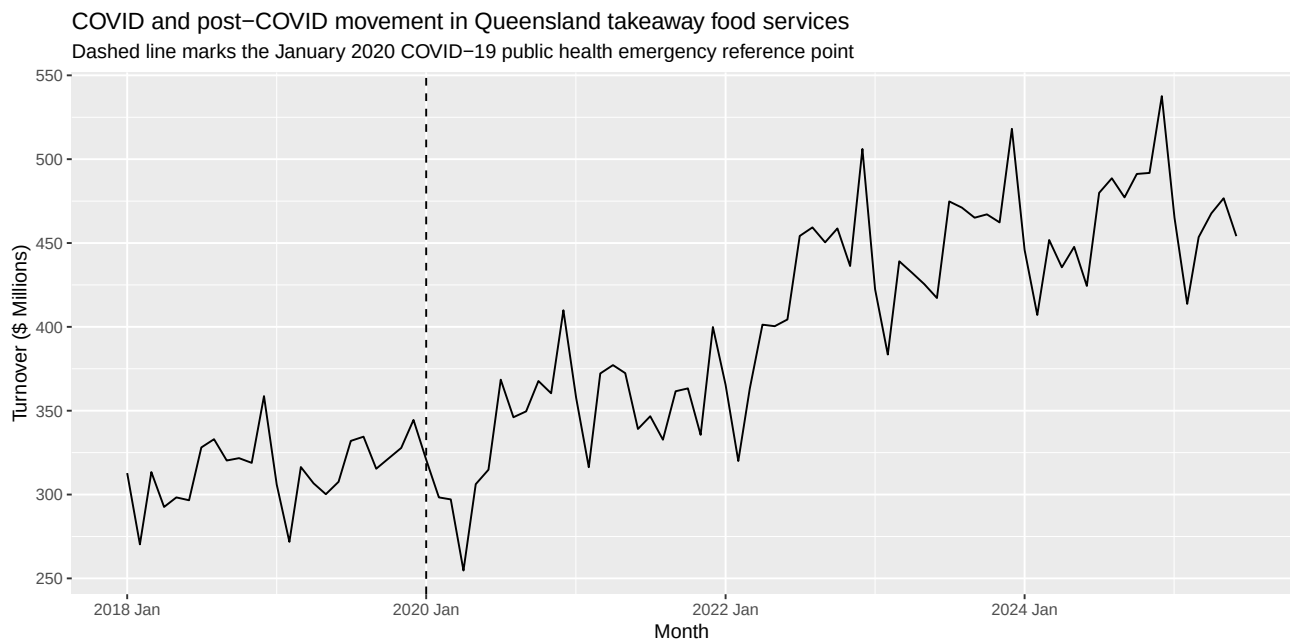


Figure 12: Queensland takeaway food services turnover from 2018 onward.

Table 8 shows that the updated series runs from April 1982 to June 2025, with one observation for each

month. Figure 11 shows that the updated data continue the long-run upward trend. Figure 12 focuses on the COVID and post-COVID period. I use January 2020 as a reference point because Queensland declared a public health emergency for COVID-19 on 29 January 2020 (The State of Queensland, 2020).

Turnover fell sharply around April 2020, but the effect appears mainly limited to that month. It recovered soon after and quickly returned to its previous level. During 2021, the series stayed relatively stable and above the pre-COVID level. From 2022 onwards, turnover increased more strongly and continued moving upward. Overall, Queensland takeaway food services appear to have been briefly disrupted by COVID, then recovered and continued growing in the post-COVID period.

Question 9

Generate forecasts for the period post 2022 until the end of your sample, from the models considered “best” in all assignments. More specifically, generate forecasts from the best benchmark, the best ETS and best ARIMA model. Plot the forecasts (both point forecasts and prediction intervals) together with the observed data and comment on these. (Make sure you can clearly visualise these. You may choose to plot on multiple graphs.) (6 marks)

```
q9_train <- updated_series %>%
  filter_index(. ~ "2022 Dec")

q9_test <- updated_series %>%
  filter_index("2023 Jan" ~ .)

q9_fit <- q9_train %>%
  model(benchmark = RW(log(y) ~ drift()),
        ets = ETS(y ~ error("M") + trend("A") + season("M")),
        arima = ARIMA(log(y) ~ pdq(2,1,0) + PDQ(0,1,1)))

q9_fc <- q9_fit %>%
  forecast(h = nrow(q9_test))

q9_fc %>%
  autoplot(updated_series %>% filter(year(Month) >= 2022),
            level = NULL) +
  geom_vline(xintercept = yearmonth("2023 Jan"),
             linetype = "dashed") +
  labs(title = "Post-2022 point forecasts compared with observed turnover",
        x = "Month",
        y = "Turnover ($ Millions)")
```

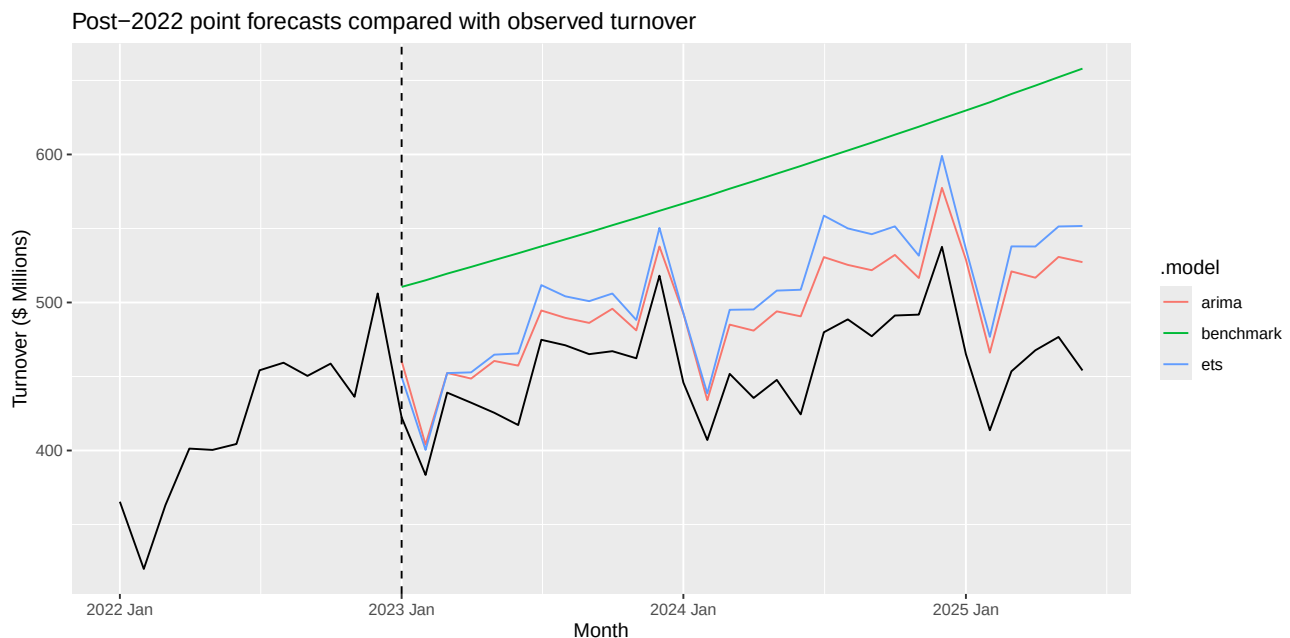


Figure 13: Post-2022 point forecasts from the best benchmark, ETS and ARIMA models.

```
q9_fc %>%
  autoplot(updated_series %>% filter(year(Month) >= 2022),
    level = c(80, 95)) +
  geom_vline(xintercept = yearmonth("2023 Jan"),
    linetype = "dashed") +
  facet_wrap(~ .model, ncol = 1) +
  labs(title = "Post-2022 forecasts compared with observed turnover",
    subtitle = "Dashed line marks the start of the forecast period",
    x = "Month",
    y = "Turnover ($ Millions)")
```

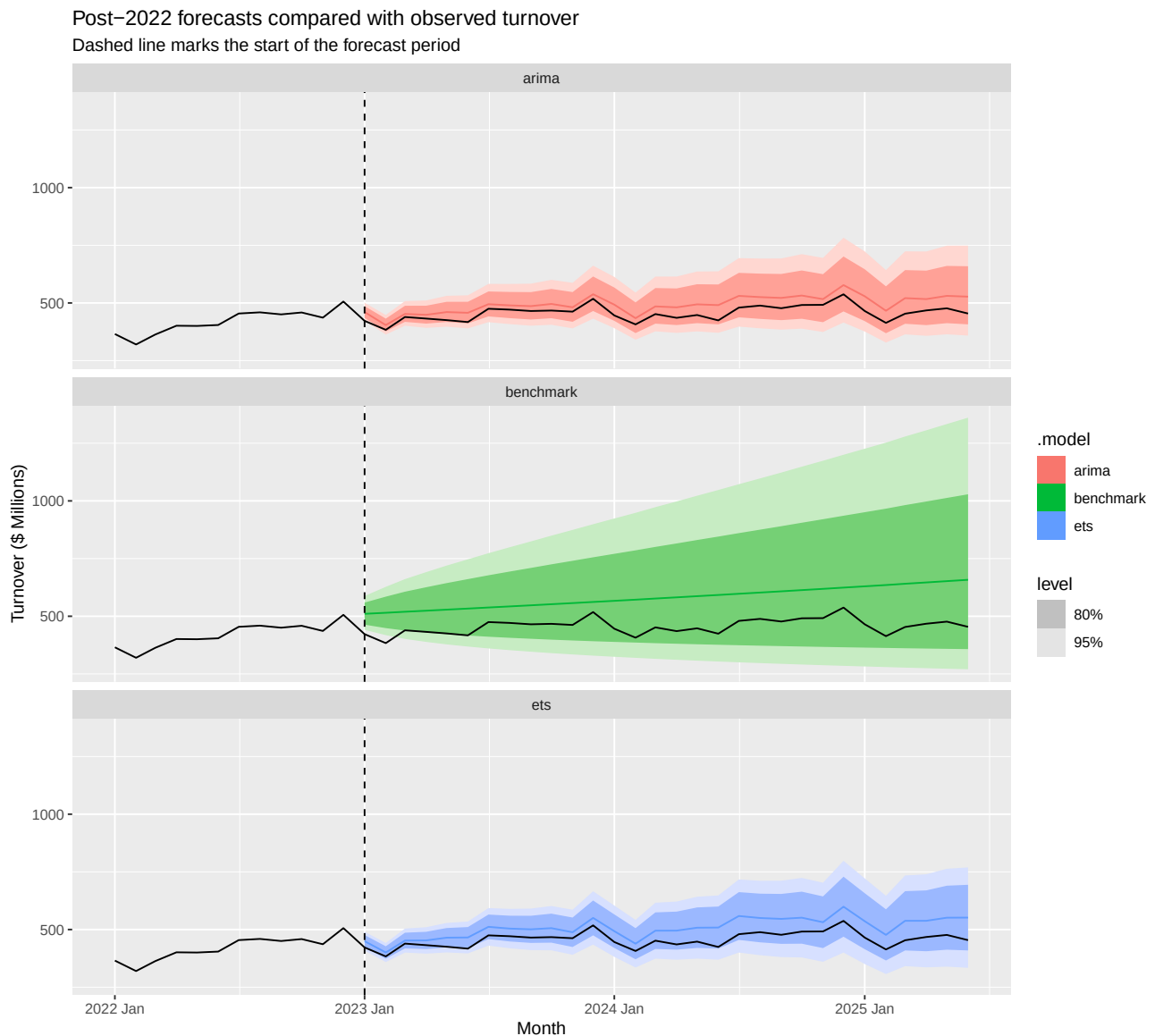


Figure 14: Post-2022 forecasts and prediction intervals from the best benchmark, ETS and ARIMA models.

```
q9_fc %>%
  filter(.model %in% c("ets", "arima")) %>%
  autoplot(updated_series %>% filter(year(Month) >= 2022),
    level = c(80, 95)) +
  geom_vline(xintercept = yearmonth("2023 Jan"),
    linetype = "dashed") +
```

```
labs(title = "Post-2022 ETS and ARIMA forecasts with prediction intervals",
     subtitle = "Dashed line marks the start of the forecast period",
     x = "Month",
     y = "Turnover ($ Millions)")
```

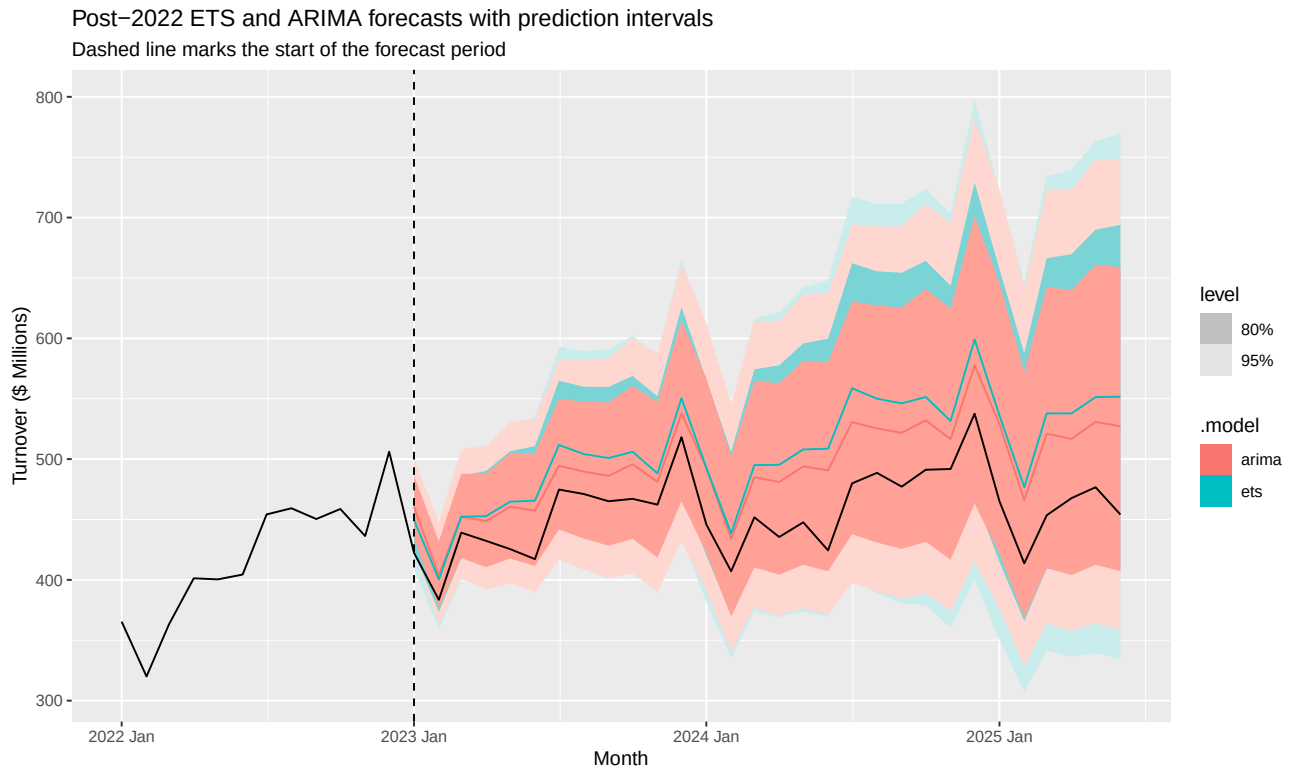


Figure 15: Post-2022 ETS and ARIMA forecasts with prediction intervals.

The models are fitted using data up to December 2022 and forecast from January 2023 to June 2025. Figure 13 overlays the point forecasts from the three models, making direct comparison easier. All three models tend to overestimate turnover in parts of the post-2022 period. The benchmark drift model is clearly the weakest because it continues the upward trend but does not capture monthly seasonality. In contrast, ETS and ARIMA capture the seasonal pattern much better.

Visually, ARIMA appears closer to the observed turnover over more of the post-2022 period, especially later in the forecast horizon. Both ETS and ARIMA still miss some sharper observed drops, such as the late-year drop around November, where actual turnover falls more strongly than the forecasts. ETS appears slightly closer at the very beginning of the forecast period, particularly around January and February 2023, where ARIMA is above the observed values.

Figure 14 shows the forecasts and prediction intervals for all three models. The benchmark has much wider intervals, reflecting greater forecast uncertainty. ETS and ARIMA show more similar seasonal forecast patterns. Figure 15 compares their intervals more directly. ETS has narrower intervals before around January 2024, but its intervals become wider than ARIMA's later in the forecast period. The January 2023 observation falls outside ARIMA's interval but is captured by ETS's 80% interval, suggesting ETS better captures uncertainty for that specific early observation. The first-year supporting check and the overall point forecast accuracy are evaluated quantitatively in Question 10.

Question 10

Evaluate the accuracy of the point forecasts over the period post 2022. A table with accuracy measures will be necessary to be presented here. Comment on which forecasts are the most accurate. (4 marks)

```
q10_accuracy <- q9_fc %>%
  accuracy(updated_series) %>%
  select(.model, ME, RMSE, MAE, MAPE, MASE, RMSSE) %>%
  arrange(RMSE)

q10_accuracy %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  kable()
```

Table 9: Post-2022 point forecast accuracy for the best benchmark, ETS and ARIMA models.

.model	ME	RMSE	MAE	MAPE	MASE	RMSSE
arima	-38.417	41.778	38.417	8.465	2.607	1.867
ets	-50.863	55.293	50.863	11.115	3.452	2.471
benchmark	-124.857	131.983	124.857	27.729	8.473	5.898

```
q9_accuracy_2023 <- q9_fc %>%
  filter_index("2023 Jan" ~ "2023 Dec") %>%
  accuracy(updated_series %>% filter_index("2023 Jan" ~ "2023 Dec")) %>%
  select(.model, ME, RMSE, MAE, MAPE) %>%
  arrange(RMSE)

q9_accuracy_2023 %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  kable()
```

Table 10: Point forecast accuracy for the first forecast year, 2023.

.model	ME	RMSE	MAE	MAPE
arima	-24.161	25.647	24.161	5.471
ets	-30.701	32.240	30.701	6.841
benchmark	-87.625	90.389	87.625	20.006

Table 9 shows that ARIMA has the lowest RMSE, MAE, MAPE, MASE and RMSSE, so it gives the most accurate point forecasts over the full post-2022 period. ETS is the second-best model, while the benchmark drift model performs worst. The supporting first-year check in Table 10 gives the same ranking using ME, RMSE, MAE and MAPE. I do not report MASE and RMSSE in the 2023-only supporting table because this table is only a short-horizon check. When only the one-year test period is supplied to `accuracy()`, these scaled measures can return `NaN`, so I use ME, RMSE, MAE and MAPE for this supplementary comparison.

The negative ME values in both tables show that all three models overestimate turnover on average. This is especially large for the benchmark model, which supports the visual pattern in Figure 13 and Figure 14. The benchmark also performs poorly because it does not capture the monthly seasonal pattern.

Question 11

Use all three models to forecast the next 24-months of your updated series. Generate the necessary plots and comment on the forecasts. Make sure you can clearly visualise these. How have your models fared amidst the effects of COVID-19? (9 marks)

```
q11_fit <- updated_series %>%  
  model(benchmark = RW(log(y) ~ drift()),  
        ets = ETS(y ~ error("M") + trend("A") + season("M")),  
        arima = ARIMA(log(y) ~ pdq(2,1,0) + PDQ(0,1,1)))  
  
q11_fc <- q11_fit %>%  
  forecast(h = "24 months")
```

```
q11_fc %>%  
  autoplot(updated_series %>% filter(year(Month) >= 2022),  
            level = NULL) +  
  geom_vline(xintercept = yearmonth("2025 Jul"),  
             linetype = "dashed") +  
  labs(title = "Twenty-four month point forecasts from the updated series",  
        subtitle = "Dashed line marks the start of the forecast period",  
        x = "Month",  
        y = "Turnover ($ Millions)")
```

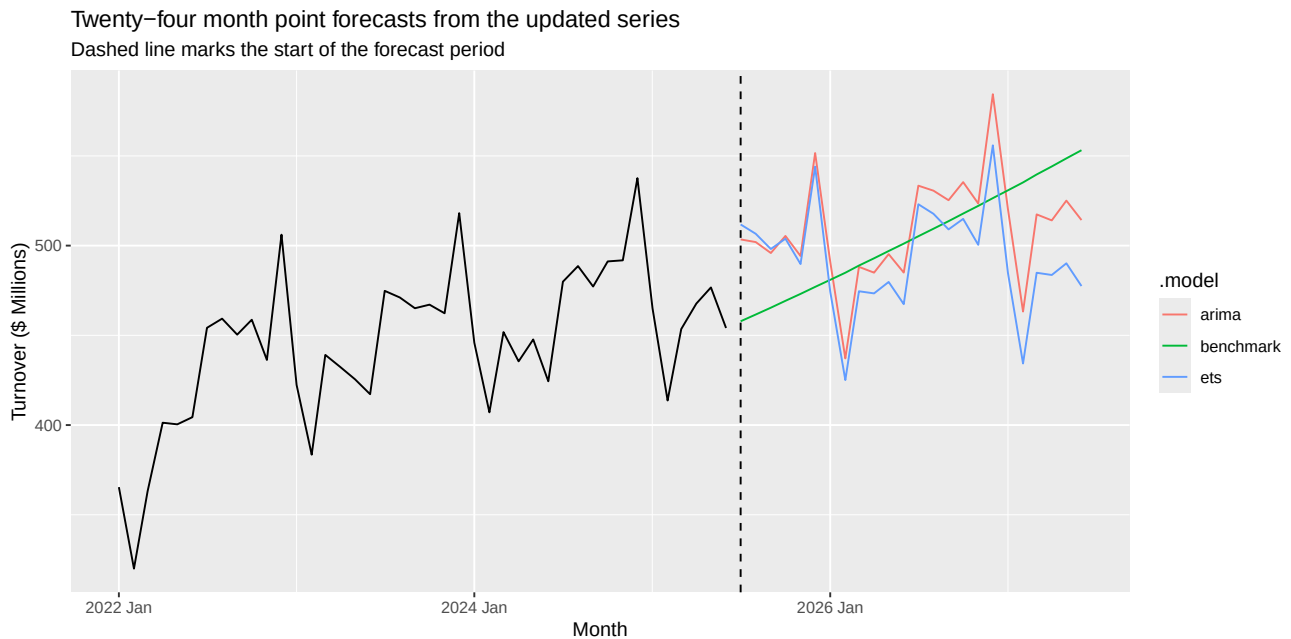


Figure 16: Twenty-four month point forecasts from the best benchmark, ETS and ARIMA models.

```
q11_fc %>%  
  autoplot(updated_series %>% filter(year(Month) >= 2022),  
            level = c(80, 95)) +  
  geom_vline(xintercept = yearmonth("2025 Jul"),  
             linetype = "dashed") +  
  facet_wrap(~ .model, ncol = 1) +  
  labs(title = "Twenty-four month forecasts from the updated series",  
        subtitle = "Dashed line marks the start of the forecast period",
```

```
x = "Month",
y = "Turnover ($ Millions)")
```

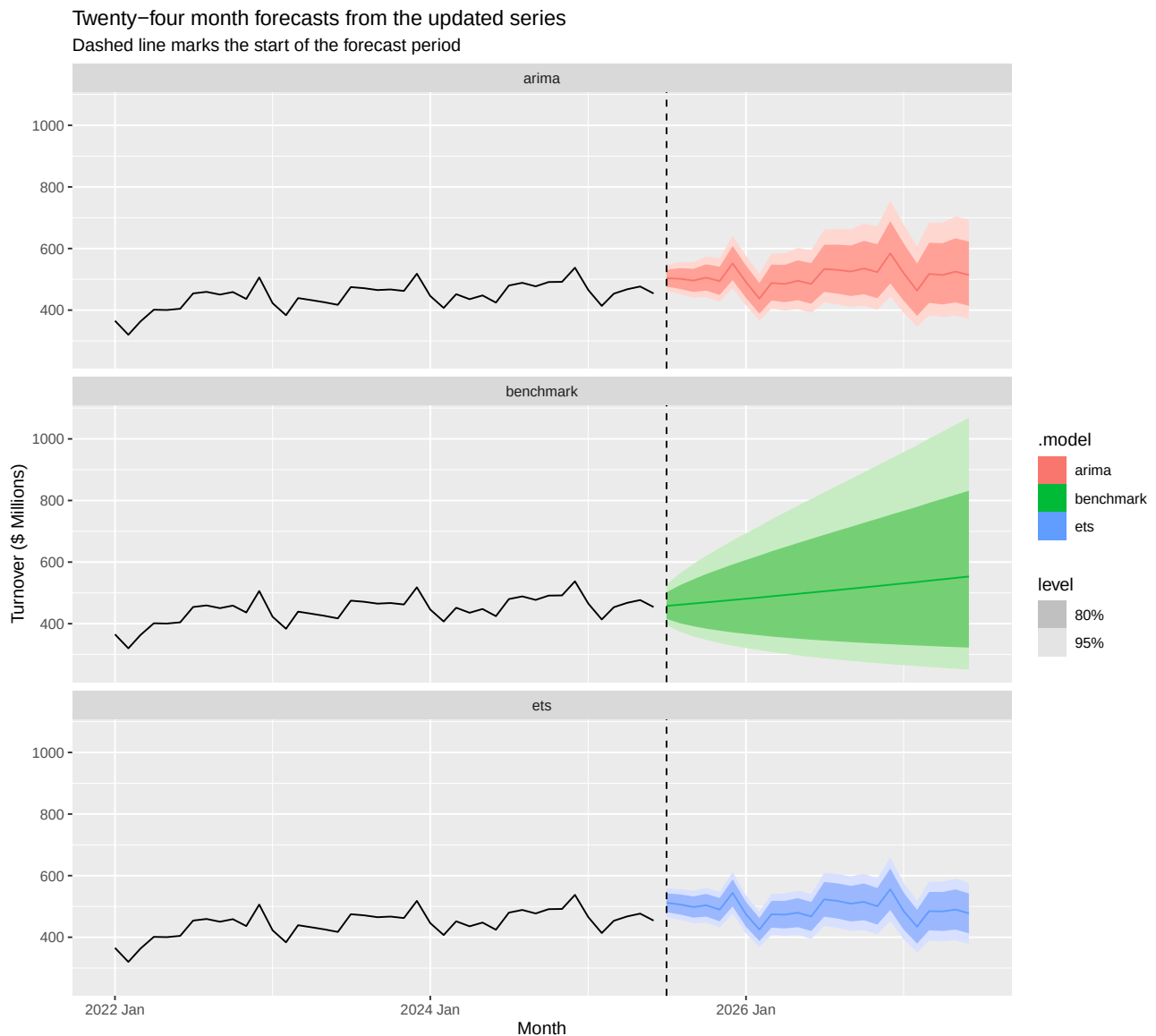


Figure 17: Twenty-four month forecasts from the best benchmark, ETS and ARIMA models fitted to the updated series.

```
q11_fc %>%
  filter(.model %in% c("ets", "arima")) %>%
  autoplot(updated_series %>% filter(year(Month) >= 2022),
    level = c(80, 95)) +
  geom_vline(xintercept = yearmonth("2025 Jul"),
    linetype = "dashed") +
  labs(title = "ETS and ARIMA twenty-four month forecasts",
    subtitle = "Dashed line marks the start of the forecast period",
    x = "Month",
    y = "Turnover ($ Millions)")
```

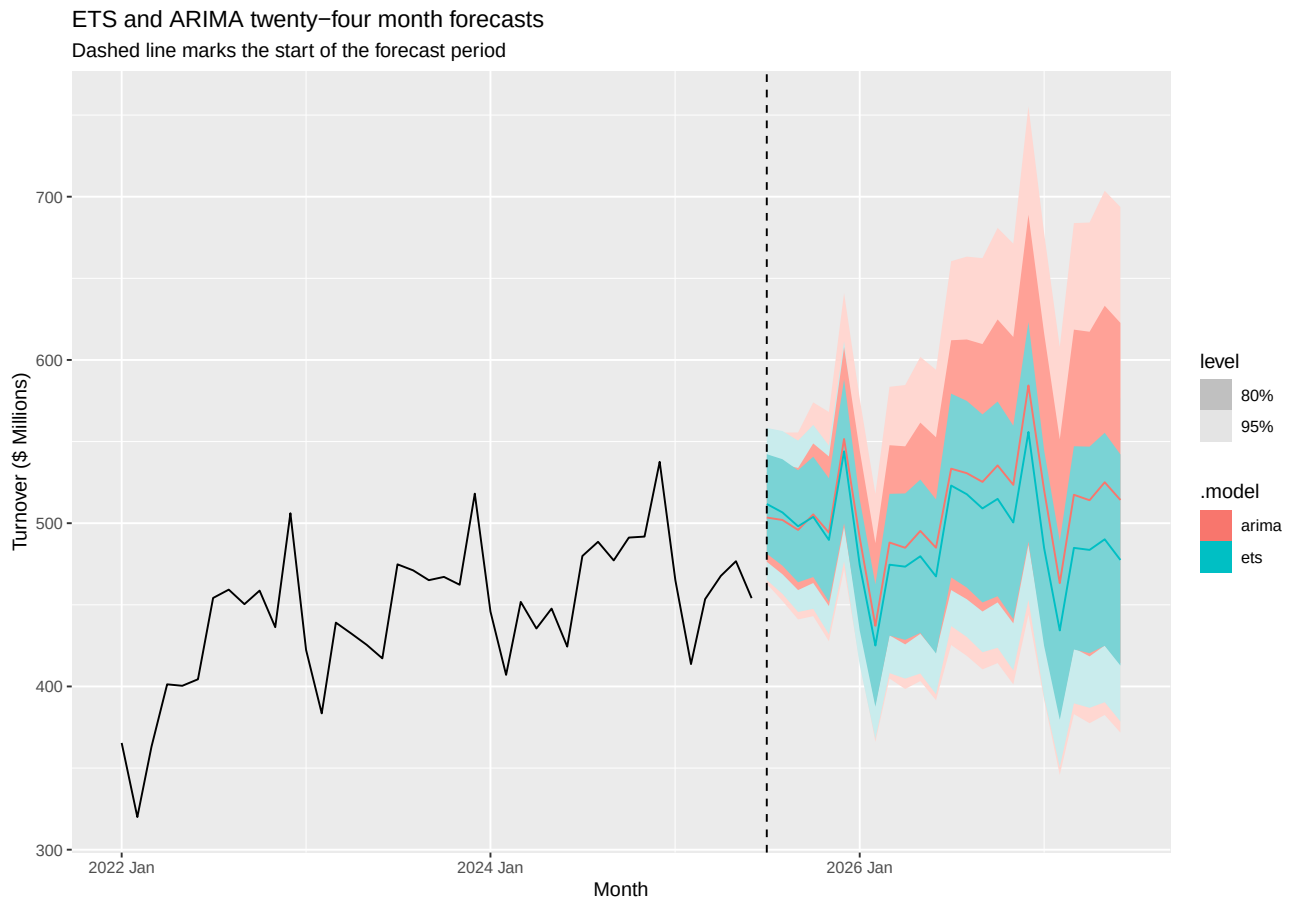



Figure 18: Twenty-four month forecasts from the ETS and ARIMA models fitted to the updated series.

Figure 16 overlays the 24-month point forecasts after refitting all three models to the full updated series. The benchmark drift model continues the upward movement but does not capture monthly seasonality. ETS and ARIMA both retain the seasonal movements and forecast turnover at the higher post-COVID level.

Figure 17 shows the corresponding prediction intervals for all three models. The benchmark has much wider intervals, reflecting accumulated uncertainty from a random walk with drift. Since the benchmark is clearly different and less useful for seasonal movement, I then compare ETS and ARIMA together in Figure 18 as a focused comparison of the two seasonal models.

In Figure 18, ETS and ARIMA have similar seasonal forecast patterns, but their uncertainty differs. ARIMA has much wider prediction intervals, especially as the forecast horizon increases, suggesting greater model-implied uncertainty. ETS gives tighter intervals, while ARIMA had the best first-year and overall post-2022 point forecast accuracy in Table 10 and Table 9. This suggests that ARIMA fared best for point forecasting, although its future forecasts are more uncertain than ETS.

The COVID disruption appears brief in Figure 12, so the models adapt reasonably well after being refitted to the updated data. However, Table 9 showed that all three models overestimated the post-2022 period on average, so the future forecasts should still be interpreted cautiously. None of the models explicitly includes a COVID intervention effect; they only learn from the updated post-COVID pattern.

Reference

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AI Declaration

I used ChatGPT for grammar correction, rephrasing, and improving the clarity of my explanations. All code, model choices, interpretations, and final edits were reviewed and decided by me.